

COMPUTER SCIENCE 423 - SECTION 1

ANALYSIS OF ALGORITHMS

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Office Hours: Monday, Wednesday 12:00- 1:30

Textbook: "Analysis of Algorithms an Active learning approach",
Jeffrey McConnel, Second edition, Jones and Bartlett
ISBN 9780763707828.

Prerequisite: CSC1 315, MATH 212 and MATH 436.

Objectives: This course introduces the techniques used to analyze the time complexities of computer algorithms and surveys algorithms used to solve common programming problems.

Topics include "big-oh" analysis, recursion, sorting algorithms, selection, graphs, priority queues and dynamic programming.

Grading: There will be two examinations, written homework assignments and programming projects. Each examination counts for 20 percent of your grade. The homework and programming projects count for the remaining 60 percent.
No Late assignment will be accepted, and no make up exams.

Attendance: Attendance is expected for all classes and is required for examinations. The student is responsible for all material missed because of absence for any reason.
Lecture Slides will be posted for one week only on BlackBoard

Summary of Topics

Introduction	1 Week
Big-Oh Analysis Basics	2 weeks
Recursive Algorithms	2 weeks
Searching and Selection Algorithms	2 weeks
Analysis of Sorting Algorithms	3 weeks
Graph Algorithms	1 Week
Dynamic Programming and Shortest Paths	1 week
NP-Completeness	1 week
Parallel Algorithms	1 Week